Supplementary Material of "Physics-Informed Model Reduction with Coriolis/Centripetal Force Terms Cancellation in 4PPa-2PaR Parallel Manipulator"

Weilong Yang, Silu Chen, *Senior Member, IEEE*, Xiangjie Kong, Chi Zhang, *Senior Member, IEEE* and Guilin Yang, *Member, IEEE*

A-I. Compound Motion of the End-effector and Decomposition-Superposition Strategy

According to the kinematic mapping established in (9a), let the generalized task-space velocity of the end-effector be denoted as $\dot{\boldsymbol{\chi}} = [\dot{x}\ \dot{y}\ \dot{z}\ \dot{\theta}]^T$. Any arbitrary 4-DOF compound motion within the reachable workspace can be mathematically decoupled into a combination of four independent principal motion vectors

$$\dot{\boldsymbol{\chi}} = \dot{\boldsymbol{\chi}}_X + \dot{\boldsymbol{\chi}}_Y + \dot{\boldsymbol{\chi}}_Z + \dot{\boldsymbol{\chi}}_\theta \tag{A1}$$

where $\dot{\boldsymbol{\chi}}_X = [\dot{x}\ 0\ 0\ 0]^T$, $\dot{\boldsymbol{\chi}}_Y = [0\ \dot{y}\ 0\ 0]^T$, $\dot{\boldsymbol{\chi}}_Z = [0\ 0\ \dot{z}\ 0]^T$, and $\dot{\boldsymbol{\chi}}_\theta = [0\ 0\ 0\ \dot{\theta}]^T$.

By applying the inverse Jacobian matrix $\boldsymbol{J}_q \triangleq \boldsymbol{J}^{-1}\boldsymbol{G}$, the corresponding active joint velocity vector $\dot{\boldsymbol{q}} = [\dot{q}_1\ \dot{q}_2\ \dot{q}_3\ \dot{q}_4]^T$ can be explicitly mapped. Based on the principle of superposition, the joint velocities induced by the compound motion are precisely the vector sum of the joint velocities induced by each orthogonal principal motion

$$\begin{aligned}\dot{\boldsymbol{q}} =& \boldsymbol{J}_q\dot{\boldsymbol{\chi}}_X + \boldsymbol{J}_q\dot{\boldsymbol{\chi}}_Y + \boldsymbol{J}_q\dot{\boldsymbol{\chi}}_Z + \boldsymbol{J}_q\dot{\boldsymbol{\chi}}_\theta \\ =& \dot{\boldsymbol{q}}_X + \dot{\boldsymbol{q}}_Y + \dot{\boldsymbol{q}}_Z + \dot{\boldsymbol{q}}_\theta\end{aligned} \tag{A2}$$

Specifically, the mapped active joint velocity components for each principal motion are expressed as follows

$$\dot{\boldsymbol{q}}_X = [\dot{q}_{1x}\ \dot{q}_{2x}\ \dot{q}_{3x}\ \dot{q}_{4x}]^T = \boldsymbol{J}_q[\dot{x}\ 0\ 0\ 0]^T \tag{A3}$$

$$\dot{\boldsymbol{q}}_Y = [\dot{q}_{1y}\ \dot{q}_{2y}\ \dot{q}_{3y}\ \dot{q}_{4y}]^T = \boldsymbol{J}_q[0\ \dot{y}\ 0\ 0]^T \tag{A4}$$

$$\dot{\boldsymbol{q}}_Z = [\dot{q}_{1z}\ \dot{q}_{2z}\ \dot{q}_{3z}\ \dot{q}_{4z}]^T = \boldsymbol{J}_q[0\ 0\ \dot{z}\ 0]^T \tag{A5}$$

$$\dot{\boldsymbol{q}}_\theta = [\dot{q}_{1\theta}\ \dot{q}_{2\theta}\ \dot{q}_{3\theta}\ \dot{q}_{4\theta}]^T = \boldsymbol{J}_q[0\ 0\ 0\ \dot{\theta}]^T \tag{A6}$$

Consequently, the velocity of the i-th active joint is the sum of its decoupled principal components, $\dot{q}_i = \dot{q}_{ix} + \dot{q}_{iy} + \dot{q}_{iz} + \dot{q}_{i\theta}$ for $i \in \{1, \ldots, 4\}$.

A-II. Algorithm 1: Calculation of Upper Bound for (28)

To ensure the full reproducibility of the exhaustive evaluation over the compact domain $IK(\Omega_c)$, the detailed sampling parameters, constraints, and the singularity exclusion rule are explicitly summarized in Algorithm 1.

Algorithm 1 Exhaustive Evaluation of the Upper Bound over $IK(\Omega_c)$

Input: Time parameters: $t_{\text{total}} = 10$ s, $dt = 5$ ms
Geometric limits: $\delta \in [0.20, 1.10]$ m, $\theta \in [-45°, 45°]$
Structural filter limits: $s_l = 0.20$ m, $s_s = 0.10$ m
Kinematic limits: $v_{\max} = 2$ m/s, $a_{\max} = 25$ m/s^2
Singularity threshold: $\text{cond}_{\max} = 10^3$
Output: Valid strict upper bound value $C_{\max}$

```
Initialize C_max ← 0
for t = 0 to t_total with step dt do
    Generate q(t), q̇(t), q̈(t) via fifth-degree polynomial interpolation
    if δ ∉ [0.20, 1.10] or θ ∉ [−45°, 45°] then
        continue {Out of defined joint space}
    end if
    if not (s_l ≤ q_1 < q_2 − 2s_s ≤ 1.75) then
        continue {Structural interference detected}
    end if
    if ||q̇||_∞ > v_max or ||q̈||_∞ > a_max then
        continue {Exceeds kinematic saturation}
    end if
    Compute the forward Jocobian matrix G using (7)
    if cond(G) > cond_max then
        continue {Near-singular neighborhood exclusion}
    end if
    Evaluate the current bounding term C_temp using (28)
    if C_temp > C_max then
        C_max ← C_temp
    end if
end for
return C_max
```

A-III. Analysis of Numerical Differentiation and Filtering Strategy

In the experimental validation, active joint accelerations are derived from encoder position data using a backward difference method. As is inherent to numerical differentiation, this process amplifies high-frequency quantization and measurement noise. To ensure high-fidelity dynamic validation, an offline low-pass filter is applied to the raw differentiated signals.

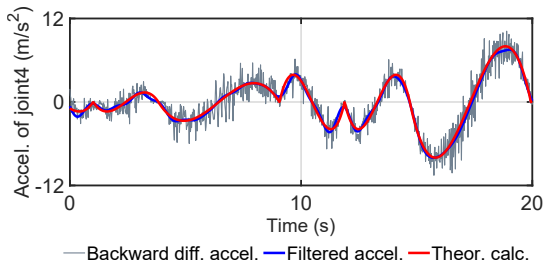

Fig. A1. Comparison of joint acceleration profiles: raw data derived via backward difference, offline low-pass filtered data, and the theoretical calculation.

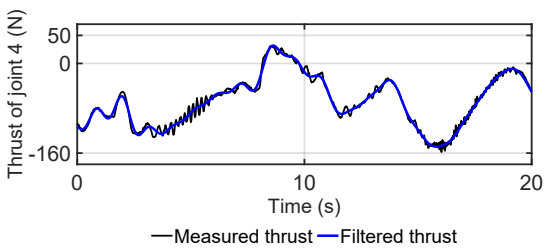

Fig. A2. Dynamic tracking performance of the motor thrust for joint 4.

Fig. A1 compares the raw acceleration computed by backward difference, the filtered acceleration, and the theoretical calculation, all of which are for the active joint 4. Quantitative analysis reveals that the root-mean-square error (RMSE) of the filtered acceleration compared to the theoretical rigid-body dynamics is tightly bounded at 0.33 $\mathrm{m/s^2}$. Furthermore, the average relative error over the entire trajectory is only 3.26 % of the full-scale amplitude. This quantitatively confirms that the high-frequency numerical errors are fully suppressed without distorting the true dynamics.

Additionally, Fig. A2 further illustrates the efficacy of this filtering approach on the measured motor thrust of joint 4. Specifically, it is observed that the filtered trajectory accurately tracks the transient peaks of the raw measured thrust without introducing temporal lag. This demonstrates that the low-pass filter effectively rejects high-frequency measurement noise while fully preserving the true dynamic amplitudes and timing of the mechanism. Quantitative analysis reveals that the RMSE is tightly bounded at 5.56 N, representing only 2.72 % of the full-scale amplitude. The high degree of consistency confirms that the error introduced by the backward-difference calculation, post-filtering, is negligible and fully within the acceptable tolerance for the proposed method, as shown in the identification results of Fig. 11.

A-IV. Geometric and Simulation Parameters of the 4PPa-2PaR PM

To fully support the kinematic modeling and dynamic simulations presented in the main manuscript, this supplementary material details the specific configuration data of the 4PPa-2PaR PM across two comprehensive tables. Table A-I defines the fundamental geometric parameters, establishing the symbolic notations for the physical structural dimensions. Furthermore, Table A-II provides the exact physical and dynamic parameters extracted directly from the CAD model. This table specifies the individual mass properties of all moving links, the principal moments of inertia, and the gravitational constant, which collectively ensure the high fidelity and strict reproducibility of the numerical evaluations and dynamic bounding limits.

TABLE A-I
Geometric parameters of the 4PPa-2PaR PM

Symbol	Description	Unit
a	Distance between revolute joints P_1 to P_2	m
b	Distance between two linear guides	m
l_u	Length of the proximal parallelogram	m
l_d	Length of the distal parallelogram	m
h_1	Distance between E_j and $\mathrm{E}_j^{'}$	m
h_2	Distance between P_2 and H_2	m
q_i	Displacement of prismatic joint along the Y axis	m
x	x coordinate of the end-effector's center point	m
y	y coordinate of the end-effector's center point	m
z	z coordinate of the end-effector's center point	m
θ	direction angle of the X axis of frame $\{E\}$ with respect to frame $\{B\}$	rad

TABLE A-II
Simulation parameters of the PM in CAD

Parameters	Values	Unit
Acceleration of gravity g	9.807	$\mathrm{m/s^2}$
Mass of the active prismatic joints m_a	2.9254	kg
Mass of the proximal parallelogram m_p	0.1320	kg
Mass of the T-shape connector m_c	0.3351	kg
Mass of the distal parallelogram m_d	0.1535	kg
Mass of the end-effector m_e	2.4718	kg
Moment of inertia I_x	3.9082	$\mathrm{g{\cdot}m^2}$
Moment of inertia I_y	6.2379	$\mathrm{g{\cdot}m^2}$
Moment of inertia I_z	1.4136	$\mathrm{g{\cdot}m^2}$

A-V. Entries in Jacobian Matrix (9)

$$k_1 = -(2\mathrm{c}\alpha_1\mathrm{c}\beta_1\mathrm{c}\beta_2\mathrm{c}\theta + \mathrm{c}\beta_1\mathrm{s}\alpha_1\mathrm{s}\beta_2\mathrm{s}\theta - \mathrm{c}\beta_2\mathrm{s}\alpha_1\mathrm{s}\beta_1\mathrm{s}\theta)/[4\mathrm{s}(\beta_1+\beta_2)\mathrm{s}\alpha_1\mathrm{c}\theta] \tag{A7a}$$

$$k_2 = -(2\mathrm{c}\alpha_2\mathrm{c}\beta_1\mathrm{c}\beta_2\mathrm{c}\theta + \mathrm{c}\beta_1\mathrm{s}\alpha_2\mathrm{s}\beta_2\mathrm{s}\theta - \mathrm{c}\beta_2\mathrm{s}\alpha_2\mathrm{s}\beta_1\mathrm{s}\theta)/[4\mathrm{s}(\beta_1+\beta_2)\mathrm{s}\alpha_2\mathrm{c}\theta] \tag{A7b}$$

$$k_3 = -(\mathrm{s}\beta_2\mathrm{c}\alpha_1\mathrm{c}\beta_1\mathrm{c}\theta + \mathrm{s}\alpha_1\mathrm{s}\beta_1\mathrm{s}\theta)/[2\mathrm{s}(\beta_1+\beta_2)\mathrm{s}\alpha_2\mathrm{c}\theta] \tag{A7c}$$

$$k_4 = -(\mathrm{s}\beta_1\mathrm{c}\alpha_2\mathrm{c}\beta_2\mathrm{c}\theta + \mathrm{s}\alpha_2\mathrm{s}\beta_2\mathrm{s}\theta)/[2\mathrm{s}(\beta_1+\beta_2)\mathrm{s}\alpha_2\mathrm{c}\theta] \tag{A7d}$$

$$k_5 = -(\mathrm{s}\alpha_1\mathrm{s}\beta_1)/(\mathrm{c}\alpha_1\mathrm{c}\beta_1) \tag{A7e}$$

$$k_6 = (\mathrm{s}\alpha_2\mathrm{s}\beta_2)/(\mathrm{c}\alpha_2\mathrm{c}\beta_2) \tag{A7f}$$

$$k_7 = -a(\mathrm{c}\alpha_1\mathrm{c}\beta_1\mathrm{c}\theta + \mathrm{s}\alpha_1\mathrm{s}\beta_1\mathrm{s}\theta)/(2\mathrm{c}\alpha_1\mathrm{c}\beta_1) \tag{A7g}$$

$$k_8 = -a(\mathrm{c}\alpha_1\mathrm{c}\beta_1\mathrm{c}\theta - \mathrm{s}\alpha_1\mathrm{s}\beta_1\mathrm{s}\theta)/(2\mathrm{c}\alpha_1\mathrm{c}\beta_1) \tag{A7h}$$

$$k_9 = a(\mathrm{c}\alpha_2\mathrm{c}\beta_2\mathrm{c}\theta - \mathrm{s}\alpha_2\mathrm{s}\beta_2\mathrm{s}\theta)/(2\mathrm{c}\alpha_2\mathrm{c}\beta_2) \tag{A7i}$$

$$k_{10} = a(\mathrm{c}\alpha_2\mathrm{c}\beta_2\mathrm{c}\theta + \mathrm{s}\alpha_2\mathrm{s}\beta_2\mathrm{s}\theta)/(2\mathrm{c}\alpha_2\mathrm{c}\beta_2). \tag{A7j}$$

A-VI. Entries in Equation (21) and (22)

$$D = \begin{cases} 8\left(4l_{\mathrm{u}}^2 - (q_1 - q_2)^2\right), & i = 1, 2 \\ 8\left(4l_{\mathrm{u}}^2 - (q_3 - q_4)^2\right), & i = 3, 4 \end{cases} \tag{A8a}$$

$$P = \begin{cases} 9m_{\mathrm{p}}l_{\mathrm{u}}^2 - 2m_{\mathrm{p}}(q_1 - q_2)^2 + 4I_x, & i = 1, 2 \\ 9m_{\mathrm{p}}l_{\mathrm{u}}^2 - 2m_{\mathrm{p}}(q_3 - q_4)^2 + 4I_x, & i = 3, 4 \end{cases} \tag{A8b}$$

$$Q = \begin{cases} 6m_{\mathrm{p}}l_{\mathrm{u}}^2 - 2m_{\mathrm{p}}(q_1 - q_2)^2 - 8I_x, & i = 1, 2 \\ 6m_{\mathrm{p}}l_{\mathrm{u}}^2 - 2m_{\mathrm{p}}(q_3 - q_4)^2 - 8I_x, & i = 3, 4 \end{cases} \tag{A8c}$$

$$R = m_{\mathrm{p}}l_{\mathrm{u}}^2 + 4I_x. \tag{A8d}$$